

Retail Decision Automation Platform

My portfolio project — starting from the business, not the dashboard

I decided early on that I didn't want this project to read like every other graduate portfolio piece in this space — "I built a sales dashboard in Power BI." That's a tool demo, not a business solution. So before I opened any BI tool, I made myself sit with the business problem first. This document is that thinking, written the way I'd actually explain it to someone hiring for a business analytics or data role.

1. Understanding the Business

1.1 The company I'm working with

I built this around a fictional company I'm calling **Meridian Retail Group** — a mid-size omnichannel retailer with physical stores and an online storefront, selling general merchandise across a few regions. I picked this profile deliberately: it's complex enough to have real cross-functional problems (finance, marketing, ops all pulling from the same data differently), but not so large that the scope becomes unmanageable for a solo project.

1.2 What's actually happening in retail right now

I didn't want to frame this project around what retail analytics looked like a few years ago, so I looked into where the industry actually is in 2026, and three things stood out to me:

- **Agentic AI has stopped being a proof-of-concept.** Industry coverage this year describes agentic AI becoming a "universal feature" of retail systems — software that doesn't just report on data but reasons over it and takes action, with a human still supervising the judgment calls.
- **The expectation has shifted from "wait for the report" to "ask and get an answer."** Some retailers are already piloting conversational analytics tools that let store associates ask a plain-language question and get an answer immediately, instead of requesting a report and waiting.
- **None of this works without a proper data foundation first.** Analysts covering the space keep coming back to the same point: most retailers still don't have a unified, real-time data foundation, and that's the actual blocker before any AI layer can be useful.

This mattered to me because it changed how I scoped the project. If I'd just built a dashboard, I'd be demonstrating a skill set that was already standard five years ago. I wanted the project to show I understand where the bar is *now* — which is data pipeline discipline first, then analytics, then a layer of automation on top.

1.3 The business problems I'm designing around

These are the patterns I kept seeing described across real retail organizations, and the ones I used to shape my own project:

- Revenue keeps growing, but profit isn't growing at the same pace — nobody's watching the gap in real time.
- The products driving the most sales aren't always the ones driving the most profit, and without margin visibility, that gets missed.
- Reporting is manual and slow, so by the time a problem is visible, it's already a few weeks old.
- The business understands *what* customers bought, but not *why*, or which customers actually matter most.

2. The Problem I'm Solving

Meridian Retail Group has plenty of transactional data but no automated way to turn it into a timely, trustworthy business decision. Decisions get made late, on incomplete information, and revenue keeps getting prioritized over profitability simply because profit isn't visible in the moment.

I phrased it this way — a decision-automation problem, not a "we need a dashboard" problem — because that's the framing I want to be able to defend in an interview.

3. Why I'm Building This Now, Not the 2021 Version of It

I wanted to be explicit with myself about what makes this different from the standard template, so I laid it out side by side:

The version I'm building

The version most portfolios still build	
A static Power BI or Tableau dashboard	A pipeline + dashboard + a natural-language query layer
Answers "what happened last month"	Answers "what's happening now, and what should I do about it"
Manually refreshed	Scheduled / event-driven automation
Person reads dashboard, decides	System surfaces the decision; I approve or act (human stays in the loop)
One BI tool, full stop	Data engineering + analytics + a small automation layer

I'm not pretending I can build a full autonomous agentic commerce system solo — that's genuinely enterprise-scale, multi-year work, and claiming otherwise would undercut my own credibility. What I can honestly build is a scaled-down but real version of the same idea: an automated pipeline, a proper analytical model, and at least one automated decision — for example, a job that flags underperforming products on its own and drafts a recommended action, rather than waiting for someone to notice.

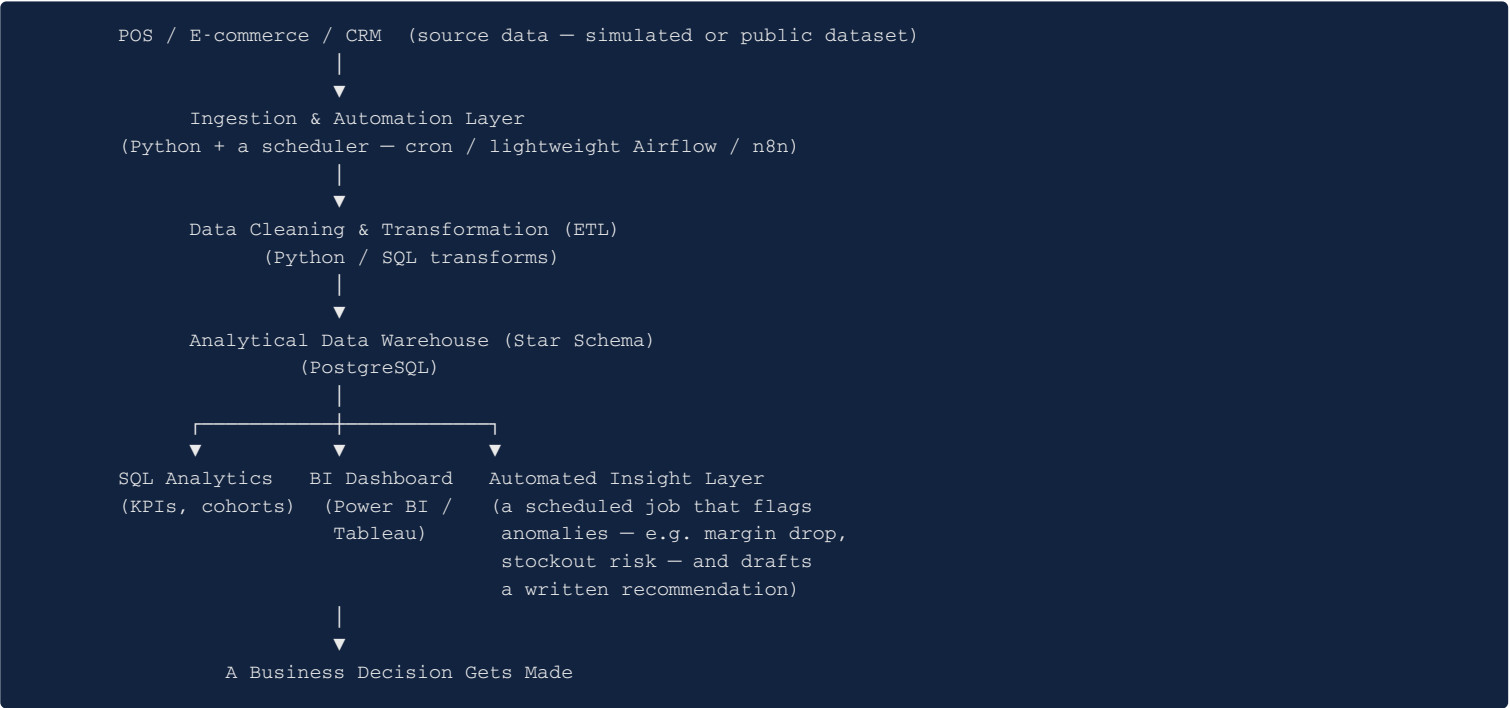
4. What I'm Actually Trying to Build

A Retail Decision Automation Platform that ingests transactional data, transforms it into a trustworthy analytical model, surfaces insight through both dashboards and natural-language queries, and automates at least one recurring business decision or alert.

5. Who I'm Building This For (Stakeholder Thinking)

Stakeholder	What they need	The question they'd actually ask
CEO	Overall business health	"Is the business actually more profitable this quarter?"
Sales Director	Product / regional performance	"Which products should I push, and which should I drop?"
Finance Manager	Margin visibility	"Where is revenue growth hiding margin erosion?"
Marketing Manager	Customer understanding	"Who are our highest-value customers, and are we keeping them?"
Operations Manager	Demand and inventory signals	"What should we be automatically reordering or flagging?"

6. How I'm Architecting It



The piece I added on purpose, the one that isn't in the standard template, is the right-hand branch. Even something as simple as a scheduled Python job that queries the warehouse, checks for a defined condition, and writes out a plain-language recommendation is enough to demonstrate the idea of decision automation without overselling what I can build solo.

7. The Data Model

Fact table — FactSales: order id, product id, customer id, date, quantity, unit price, discount, cost, revenue, profit.

Dimension tables:

- **DimCustomer** — demographics, segment, region
- **DimProduct** — category, cost, margin band
- **DimDate** — day / week / month / quarter, for trend analysis
- **DimLocation** — store, region, country

For data, I'm planning to use a public retail dataset (something like Kaggle's "Online Retail" or "Superstore" sets) or generate synthetic data myself with Python (faker + numpy) if I want more control over realistic patterns — seasonal demand spikes, discount-driven margin erosion, that kind of thing.

8. My Build Plan

Phase	What I'm doing	What comes out of it
1	Business framing + sourcing/simulating data	Problem statement, data dictionary
2	Cleaning and ETL in Python	A reproducible cleaning script/notebook
3	PostgreSQL warehouse + SQL analysis	Star schema, core business queries
4	BI dashboards	Executive, Sales, Product, Customer views
5	Automation layer	Scheduled job + one automated alert/recommendation
6	Packaging	README, architecture diagram, short demo, write-up

Realistically, working on this part-time, I'm giving myself 3-5 weeks, with extra time if I want to add forecasting or polish the dashboard visuals further.

9. What I'll Actually Have to Show

- **Code:** ETL scripts, SQL schema and queries, the automation script — all documented on GitHub
- **Dashboards:** a Power BI or Tableau file, published or screen-recorded if I don't have hosting
- **The automation artifact itself:** the actual alert or recommendation output, not just a description of what it would do
- **This write-up:** the problem, the "why now," the architecture, and the business impact
- **A short demo:** a 2-3 minute walkthrough for my portfolio site and LinkedIn

10. How I'll Talk About It

What I want to avoid saying:

"I built a retail dashboard using Power BI and SQL."

What I actually want to say:

"I designed a retail decision-automation pipeline — from raw transactional data through a warehouse to both BI dashboards and an automated alerting layer — to show how a mid-size retailer could move from monthly manual reporting to near-real-time, action-oriented analytics, in line with where retail is actually heading in 2026 with agentic and automated decision support."

That's the version of the story that shows business understanding, technical range across data engineering, analytics, and automation, and awareness of where the industry actually is — not a frozen, several-years-old skill set.

Context sourced from: Chain Store Age ("Three retail tech trends to watch in 2026 — a mid-year update"), BizTech Magazine ("AI Agents, Data Governance and Workforce Shifts Redefine Retail in 2026"), Retail Customer Experience ("2026 will be the year of agentic AI"), and Mordor Intelligence's Agentic AI in Retail and eCommerce market report.